

Standard Input / Output

Types of Input / Output

1. Console I/O
2. File I/O
3. Network I/O
4. Memory I/O

Console I/O: Jo bhi hamari screen pr print hota ha use console khte ha.

System.out.println("Console Output");

System.err.println("Console Error"); to print error statements.(good Practice).

```
public class SystemINxOP {
    public static void main(String[] args) {
        // System.out.println defination:
        System.out.println("Hello");
    }
}
/*
    java.lang package:---> ye defaultly import ho jata ha, main class mein
class System{
    public static final PrintStream out;
}

    java.io package:
class PrintStream{
    public void println(String s){
        code for printing
    }
}
*/
```

- **println**→ new line.
- **Print**→ without new line.
- **Printf**→ print formatted output.

system.in- to read data from keyboard.

```
    java.lang package:---> ye defaultly import ho jata ha, main class mein
class System{
    public static final InputStream in;
}

    java.io package:
class InputStream{
    public void read(String s){
        Reads input
    }
}
```

Keyboard input before Scanner class.

Phle hum java ki jagah ye use krte the→

keyboard→OS Buffer→Input Stream→ Input stream reader→Java Buffer →readline
methos→Program

(buffer is type of arrays that stores characters.)

1. Take Input from keyboard "Priyansh"

2. OS Buffer (23,243,354,5,334,34,65,6,)takes in integer due to ASCII, UNICODE.
3. (InputStream object) receives bytes from OS.
4. InputStreamReader: converts stream of bytes into stream of char.(P,r,i,y,a,n,s,h)
5. Buffered Reader: will read all character inputs, and stores in java buffer, in string format.
6. Program: jab bhi java ko zarurat hogi to wo Buffered reader se le-lenga, by readline();

Why Scanner class

- Simplify (input)
- Read without typecasting all → int, double, boolean, string.

Scanner

- Scanner class comes in (java.util) Package, it means it is a utility class.
- Scanner ka object apne constructor ke parameter me InputStream([System.in](#)) expect krta ha.
- constructor ka parameter decide krte ha ki Scanner ky input lega.
- Example Input from keyboard→[System.in](#), Input from file→new file(file name), etc.
- Scanner internally works on tokenization basis.
- Scanner class is slower than Buffer reader .
- For more optimized code we use buffer reader, example→ competitive programming.

Methods of SCANNER:

- next→ It only reads a word.
- nextLine→ It reads whole line, jabtak hum enter press nhi kr dete.
- nextInt→ it reads integer values.
- nextDouble→it reads double values.
- Etc, all type of datatypes.

```
import java.util.Scanner;

public class ScannerClass {
    public static void main(String[] args) {
        Scanner sc = new Scanner (System.in); // new scanner object with parameter of
Keyboard input.
        int age = sc.nextInt(); // methods of scanner.
        double num = sc.nextDouble();
        String name = sc.next();
        String college = sc.nextLine();
    }
}
```